Introduction to Artificial Intelligence

Prof. Dr. Hamed Shariat Yazdi, Darmstadt University of Applied Sciences

Assignment:
Machine learning – Classification

In this lab session, you will practice ML classification with the Kaggle challenge "Titanic: Machine learning from disaster" https://www.kaggle.com/c/titanic . You will be working in your teams of two students each.

1. Read Sections 2.1-2.4 of the book "Applied Artificial Intelligence – An Engineering Approach". Focus on the paragraphs relevant for this course. You don't need to read everything.

2. Register in Kaggle https://www.kaggle.com

3. Download training data, test data and sample submission from https://www.kaggle.com/c/titanic/data

4. Analyze the dataset.

5. Work in the PyCharm project for your team, should be named as

   - <Day and Block of the Lab>-Team-<your team number>

     - E.g. Mo3y-Team-02

   and should be committed to the Git repository of your team).
   Add a new Python package "03a_ML_classification_Titanic" and add a new Juypter notebook "Kaggle_Titanic.ipynb in this package

6. Copy the data files to a subfolder "data"

7. Implement the Kaggle challenge

8. Submit the submission file generated by your implementation to Kaggle

9. Check your position in the leaderboard.
   You may improve your position by modifying your implementation (e.g., pre-processing, other ML approaches, other ML parameters etc.) and re-submitting.

10. Commit your code to the Git repository

11. Add your team number, Kaggle score (float) and Kaggle position (int) in the shared spreadsheet of the assignment (link in Moodle).